



Western Cape
Government

Health



The South African Triage Scale (SATS)



Training manual 2012

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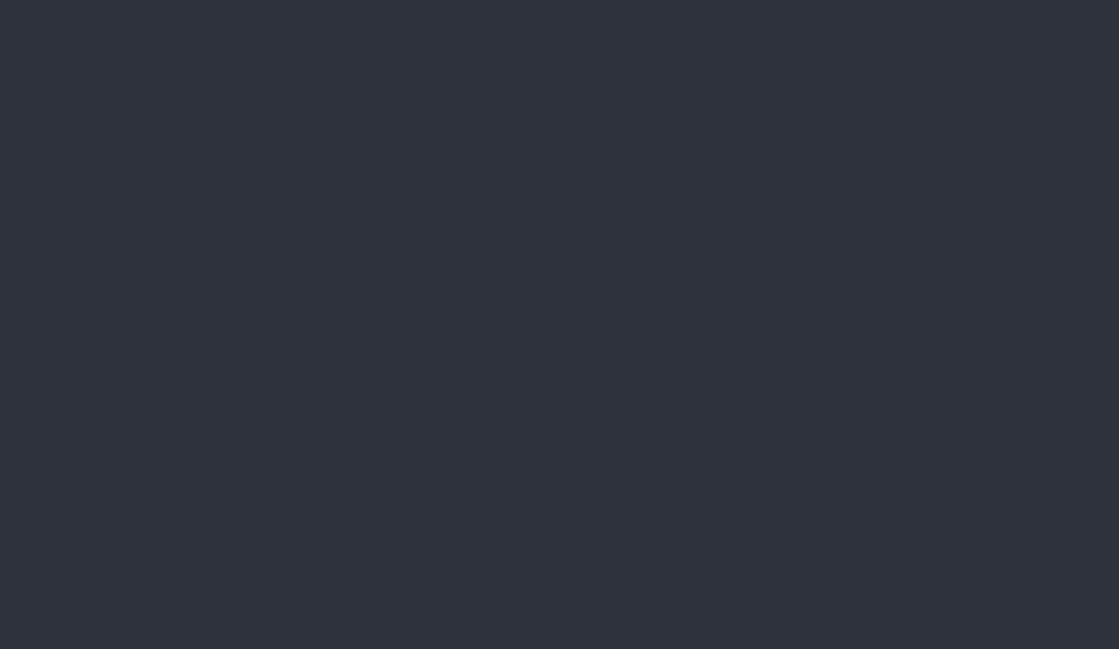
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INTRODUCTION

Learning Objectives:

- Understand the purpose of triage and the benefits of implementing the SATS
- Be familiar with the requirements for standardised triage implementation
- Understand the terminology and key concepts around triage

A nine-month-old baby boy is carried into the children's section of the outpatient department in his mother's arms. He appears to be asleep. At the triage desk he is seen by a nurse and found to have lips and tongue that are grey/blue in colour, and he is taken straight into the resuscitation room as an emergency.

In the resuscitation room he is given oxygen at 15 litres/minute by face mask with a non-rebreather reservoir bag. He is noted to be grunting and breathing very fast. His hands are cold to touch and the capillary refill time is prolonged to four seconds. An intravenous cannula is placed. A blood sample is taken at the same time for blood glucose, full blood count and blood culture. An intravenous infusion of normal saline is commenced at 20ml/kg to run as fast as it can go.

Other treatments are given, depending on the result of the investigations and the response to the treatment he receives. It is now 18 minutes since the baby came through the outpatient department's door, and his situation is stable. It is now time to take a full history and carry out a full examination to make a definitive diagnosis. He is diagnosed as having very severe pneumonia, and receives specific treatment for this. However, before coming to this diagnosis, no time was wasted, his status was stabilized, based on a few leading signs and symptoms, even when the medical staff did not know exactly what was wrong with him.

This was good triage and emergency management. Would it have happened like this in your hospital? In this training course, you are going to acquire the necessary knowledge for the correct triage of sick children and adults.

Many deaths in hospital occur within 24 hours of admission. Some of these deaths can be prevented if very sick patients (especially children) are quickly identified on their arrival and treatment is started without delay. In many hospitals around the world, children are not checked before a senior health worker examines them. As a result, some seriously ill patients have to wait a very long time before they are seen and treated. Children are known to have died of a treatable condition when waiting in the queue for their turn. The idea of triage is to prevent this from happening.

The purpose of triage is to prioritise patients based on medical urgency in contexts where there is a mismatch between demand and capacity (i.e. patient load overwhelms the available resources).

1.1 The benefits of implementing SATS

1. expedite the delivery of time-critical treatment for patients with life-threatening conditions.
2. ensure that all patients are appropriately prioritised according to their medical urgency.
3. improve patient flow.
4. improve patient satisfaction.
5. decrease the patient's overall length of stay.
6. facilitate streaming of less urgent patients.
7. provide a user-friendly tool for all levels of health care professionals.

By introducing the SATS at a public urban hospital in Cape Town, mean waiting times were reduced significantly for all priority levels except the non-urgent green category. The most dramatic reduction in waiting times was seen in patients coded as red (82%).¹

On behalf of the
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The SATG and PTWG wishes to acknowledge the WHO Department of Child and Adolescent Health and Development and Professor Elizabeth Molyneux, who developed the ETAT training course on which these course materials are based. We are also grateful to the ETAT-South Africa (ETAT-SA) working group for all their efforts and hard work in adapting and developing the generic ETAT materials in to the ETAT-SA materials specifically for the South African setting.

All reasonable precautions have been taken by the PTWG, the ETAT-SA working group, WHO and SATS group to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either express or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall the PTWG, the ETAT-SA working group, WHO or the SATG be liable for damages arising from its use.

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1.4 Terminology and key concepts

1. **Triage**, from the French word “trier”, literally means: “to sort”. The aim is to bring “the greatest good to the greatest number of people” – this is achieved through prioritising limited resources to achieve the greatest possible benefit. Patients are sorted with a scientific triage scale in order of urgency – the end result is that the patient with the greatest need is helped first.
2. **Patient to triage**: for the hospital or clinic context this refers to a patient that appears relatively stable and is able to mobilise him/herself to the designated triage area. This will be the type of triage used for most hospital and clinic cases.
3. **Triage to patient**: here the patient is usually unstable. The patient is unable to mobilise him/herself to the designated triage area and will need to be triaged where they are found. They may need to be referred directly to the resuscitation (resus) area if they are at a health facility. Triage may also be performed at the bedside and documented in retrospect. This type of triage will be used less often in the hospital context and predominantly in the pre-hospital context.
4. **Physiology** (i.e. vital signs): refers to the normal functioning of the different body systems. Some of the physiology can be readily measured (e.g. pulse, blood pressure, respiratory rate, temperature).
5. **TWES: Triage Early Warning Score**. This is a composite score of the patient’s physiology. The score is derived by assigning a number between 0 and 2 for each of the patient’s vital signs. The higher the score the greater the urgency.
6. **Streaming**: the use of dedicated healthcare resources for each priority group of patients. For green patients, this may be a doctor or nurse practitioner; this person needs their own space to see these patients.
7. **Pain**: Severe pain is unbearable, the worst pain the patient has ever felt. It may be associated with sweating, paleness, and altered level of consciousness. Moderate pain is intense, but bearable. Mild pain is any other pain. Remember to do a pain assessment on every patient that you see.
8.  **Additional investigation**: You will find this exclamation icon in later chapters. It represents an additional investigation which may lead to a change in the patient’s triage priority level. Checking the blood glucose concentration or measuring the oxygen saturation level are examples of additional investigations. Section six outlines all key additional investigations important at triage. They can also be found on the SATS charts.
9.  **Warning**: The lightbulb icon indicates a warning that usually follows immediately after an additional investigation, implying that some immediate action is required (e.g. a child whose oxygen saturation levels are found to be 80% requires oxygen administration and should be taken to the resuscitation area).
10.  **Additional tasks**: The icon with a red cross represents additional tasks that are beneficial to the patient if initiated at triage. These additional tasks do not change the patient’s triage priority level. Examples include starting oral rehydration therapy for a child that is dehydrated, cooling a burn that occurred within 3 hours or applying direct pressure to an uncontrolled haemorrhage.

band comprehensively implement the equipment needed and checklist of requirements.

	Additional equipment
Pulse oximeter with paediatric probes	ECG
	Finger prick machine, haemoglobin and glucometer measurement
Urine collection containers, urine dipsticks & urine pregnancy tests	

vide in the countries of North America, at of Emergency Medicine as a medical practitioners and nurses per unit countries. Given the significantly lower re nurse triage is widely practiced, it is be a priority in our setting.

100 population/ year	Doctor: Nurse ratio
71.2	1 : 8.0
997	1 : 4.0
330	1 : 3.4
513	1 : 1.6
479	1 : 3.0

ammum for selected countries

the Emergency Centre in most and Nursing Assistants (ENAs) have been tional standards of nursing triage.²³ by all health care professionals be the medical officer, the registered program is to empower the individual through practice and repetition that a are therefore encouraged to participate e and up to date.

1.2 Triage requirements

Triage is simple to do, but in order to standardise the process SATS as a validated tool certain requirements need to be met for the process and Appendix A on page 4 includes a detailed

Location	Equipment
Privacy: Screen, partition or separate room.	Gloves, face masks & other barrier protective devices
Safety: Security/protected	Wall clock
Size of area: pushchairs, wheelchairs, stretchers	Low reading electronic/mercury thermometer
Accessibility	Vital signs monitor OR baby nometer with paediatric
Baby-changing facilities	Dry dressings/ bandages

Table 1: Requirements for adequate / efficient triage

1.3 Who should be the triage provider?

Nurse-based triage has been successfully implemented worldwide, the Middle East and Australasia since the development of speciality about 30 years ago. Table 2 shows the number of medical staff in South Africa, compared to some "developed" countries. The doctor:nurse ratio in South Africa compared to countries where it is apparent that the development of nurse-based triage should

Country	Rate per 100,000	
	Doctors	Nurses
South Africa	56.3	4
Canada	229	8
Australia	240	8
Israel	385	6
UK	164	2

Table 2: Doctor and nurse rates per 100,000 population per year

Nurses are the first medical contact for the patients attending triage instances. In South African studies, adequately trained triage nurses have been shown to be accurate to a degree comparable with internal medicine. The South African Triage Scale should be known and applied by all triage nurses involved in the Emergency Centre. The triage provider can be a nurse, enrolled nurse or the ENA. The purpose of this triaging process is to ensure that the patient is seen by a triage provider who participates with the knowledge to triage. It will only be when the triage provider will become skilled with triage. Successful providers in triaging as frequently as possible in order to stay in practice

Patients

Criteria:	Yes	No
s, or in easy view of		
should be able to relative or carer)?		

5

ing:	Yes	No
e area to measure		
ge area?		
source of info?		
waiting area?		
ality of those triaged?		

Assessment Questions

Clearly indicate whether the following statements are true or false:

1. The purpose of triage is to prevent deterioration or death of a patient while waiting in the queue for their turn.

☐ True ☐ False
2. The triage method should be known and applied by clinical nurse practitioners only.

☐ True ☐ False
3. Streaming is the process of getting patients to wait for as long as possible.

☐ True ☐ False

Choose the correct answer:

4. The benefits of triage are:

(a) To facilitate streaming of green patients

(b) To decrease the waiting time of life-threatening conditions

(c) To ensure that patients are appropriately prioritized by urgency

(d) To prevent children from dying in the waiting room

(e) All of the above
5. TEWS is short for:

(a) Triage Early Warning System

(b) Trauma Early Warning Scale

(c) Triage Emergency Warning System

(d) Triage Early Warning Score

(e) Trauma Emergency Waiting Score

APPENDIX A: Checklist of triage emergency centre requirements

Structural Requirements

Does the triage area meet the following criteria?	
1	Is the triage area a dedicated space?
2	Is the triage area well signed?
3	Is the triage area secure (i.e. behind the security gate security staff)?
4	Is the triage area at least 10 square meters in size (i.e. accommodate a nurse, patient in a wheelchair and security staff)?

Checklist of triage Infrastructure Requirements

Content Requirements	
Does the triage area contain the following?	
1	A desk and chair?
2	Triage paperwork for adult, children and infants?
3	A wall clock with a second hand?
4	A stethoscope?
5	A low reading thermometer?
6	Dry dressings and bandages?
7	Gloves?
8	Sphygmomanometer (manual, digital or electronic)?
9	Blood glucose monitor?
10	A measuring tape OR marks displayed on wall in triage area for children (i.e one mark at 95cm and one at 150 cm)?
11	2 x different SATS posters prominently displayed in triage area?
12	SATS manual readily available for triage office as a security staff reference?
13	SATS patient info leaflet prominently displayed in the triage area?
14	Triage register or computer with register?
15	White board to track and communicate to other staff a patient's status?

2.4 Terminology and key concepts

1. **Red vs Resus:** Patients may be triaged Red on the basis of their presentation, but not necessarily be a full resuscitation case. Conversely, if a patient presents to you as a resus you do not need a triage tool to tell you that they are a Red case. For those patients who present like this (e.g. Cardiac arrest), triage before treatment is not necessary – if a patient is a resus, they are Red by definition.
2. **Majors vs Minors:** The majors area in a hospital is staffed by appropriately trained personnel and advanced equipment to deal with emergency, very urgent and urgent patients. The minors area is staffed by appropriately trained personnel and the respective equipment and resources to deal with routine or non-urgent patients.

Priority COLOUR	Target time	Management
RED	IMMEDIATE	Take to the resuscitation room for emergency management
ORANGE	< 10 mins	Refer to majors for very urgent management
YELLOW	< 1 hour	Refer to majors for urgent management
GREEN	< 4 hours	Refer to designated area for non-urgent cases
BLUE	< 2 hours	Refer to doctor for certification

Table 3: SATS priority levels and target times to be seen with-in

Assessment Questions

Clearly indicate whether the following statements are true or false:

1. There are two SATS charts – one for paediatric patients and one for adult patients.
☐ True ☐ False
2. If an emergency sign is identified in the first step, the patient is taken to the resuscitation area immediately.
☐ True ☐ False
3. If no emergency signs are identified in step one, but an urgent sign is identified in step two, the patient is immediately triaged yellow and asked to wait.
☐ True ☐ False

Choose the correct answer:

4. The SATS priority level Orange is defined as:
(a) Emergency requiring immediate intervention
(b) Very Urgent requiring intervention within 10 minutes
(c) Urgent requires intervention within 60 minutes
(d) Routine requiring intervention within 240 minutes
(e) Life threatening but not requiring any intervention
5. The senior healthcare professional's discretion refers to:
(a) The junior nurse overriding the final triage decision
(b) The clinical nurse practitioner overriding the final triage decision
(c) The medical student overriding the final triage decision
(d) The medical officer overriding the final triage decision
(e) b and d above

Flowchart

presenting complaint

TEWS

patient as to the reason for coming to the ED and the triage process already commences with the **emergency clinical signs**. The Airway, Breathing, Circulation (ABC-c-c-DO) approach is used for all patients. If a patient is assigned a Red priority level

Very Urgent or **Urgent** clinical signs.

If the TEWS is calculated, key additional

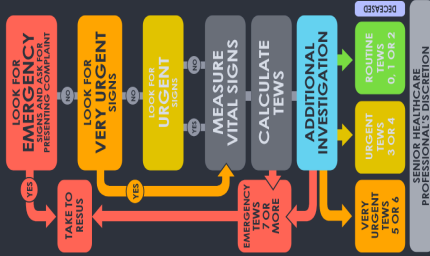


Figure 2: The SATS process flowchart

THE SATS PROCESS FLOWCHART

Learning Objectives:

- Understand the five step approach and SATS process flowchart
- Be familiar with the two versions of the SATS chart
- Be familiar with the priority levels of SATS

2.1 The five step approach

- Step 1:** Look for emergency signs and ask for the emergency centre.
- Step 2:** Look for very urgent OR urgent signs
- Step 3:** Measure the vital signs and calculate the triage priority
- Step 4:** Check key additional investigations
- Step 5:** Assign final triage priority level

Figure 1: SATS five step approach

The process of triage starts with a question to the mother/carer/ambulance crew: 'Is this child/infant/child in need of medical attention?' If the answer is 'yes', the triage practitioner rapidly assesses the patient for any of the following signs: Breathing, Circulation, Coma, Convulsion, Dehydration, Other paediatric patients. If emergency clinical signs are found, the patient is taken straight to the resuscitation area without delay.

If no Emergency clinical signs are present then check for any other signs. Whether these are present or not, vital signs are measured, the patient is reassessed and the triage priority adjusted as shown in Figure 2.

It is important to note that if a patient has any emergency signs, then a TEWS does NOT need to be calculated at triage. There should be no delay in taking the patient to the resuscitation area. Finally the senior healthcare professional's (SHCP) discretion as to whether the patient should be taken to the resuscitation area or not, overrides the final triage priority assigned.

2.2 The SATS process flowchart

There are two versions of the SATS chart as seen in Figure 4 and Figure 5. The paediatric version of the SATS chart is used to triage all paediatric patients younger than 12 years and smaller than 150 cm. The adult version of the SATS chart is used to triage patients older than 12 years or taller than 150 cm.

Both age specific versions have the exact same SATS process flowchart as depicted in Figure 2. The five steps in Figure 1 are integrated into this process flowchart. The differences are found in the next two sections describe in detail the adult and paediatric signs respectively.

2.3 SATS priority levels

The South African Triage Scale has four priority levels as shown in Table 3. Each priority level should ideally be managed within target time to treatment.





Adult SATS Chart



EMERGENCY
Obstructed Airway - not breathing
Seizure- current
Burn - facial / inhalation
Hypoglycaemia - glucose less than 3
VERY URGENT
High energy transfer (severe mechanism of injury)
Shortness of breath - acute
Level of consciousness reduced / confused
Coughing blood
Chest pain
Stabbed neck
Haemorrhage - uncontrolled (arterial bleed)
Seizure- post ictal
Focal neurology - acute (stroke)
Assault



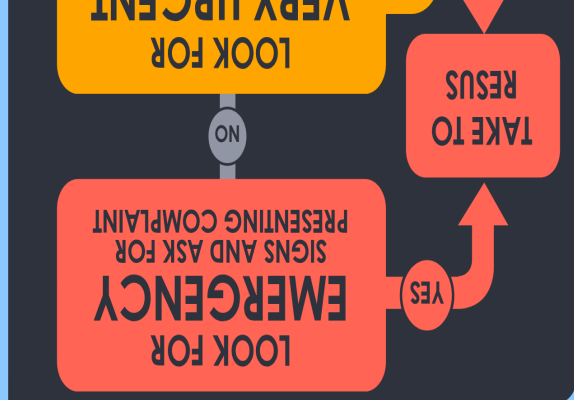




Paediatric SATS Chart



EMERGENCY	
Always and Breathing	Not breathing or reported apnoea Obstructed breathing Central cyanosis or SpO ₂ less than 92% Respiratory distress (fever)
Circulation	Cold hands + 2 or more of the following: • pulse weak and fast • capillary refill time 3 sec or more • mottling Uncontrolled bleeding (not nose bleed)
Convulsions	Convulsing or immediately post-ictal and not alert
Coma	AVPU: Responds only to Pain (P) OR Unresponsive (U) Coma
Dehydration	Diarrhoea +2 or more of the following: • skin pinch very slow - 2 sec or more • lethargy / floppy infant • very sunken eyes
Other	Facial / injection burn Hypoglycaemia recorded at any time Glucose less than 3 mmol/L Bulimic



Level of consciousness reduced / confused

Any patient that is not fully alert (i.e. confused, unresponsive to a verbal stimulus, painful stimulus or unresponsive).

Threatened limb

A patient presenting with a painful, pale, pulseless, weak, numb limb.

A Is the patient **Alert**? If not, is the patient responding to **Voice**? If not, is the patient responding to **Pain**?
P Is the patient who is **Unresponsive** to voice stimulus or **Unconscious**.
U **AND** to pain is **Unconscious**.

A threatened limb presents as:

P Pain **P** Pulselessness **P** Capillary Refill Delay
P Pallor **P** Paralysis or pins & needles **P** Temperature

3.3 URGENT SIGNS

Haemorrhage - controlled

This refers to a situation where a patient presents with an active bleed and you as the triage provider apply direct pressure with a dry dressing and are able to control the bleed. This does not refer to a patient presenting with dry blood.

Abdominal pain

In all females of child-bearing age additional investigations (i.e. urine dipstick and urine pregnancy test) should be performed to pick up a possible ectopic pregnancy.

Assessment Questions

Clearly indicate whether the following statements are true or false:

- 1. Singed facial hairs and soot around the nose and mouth are an indication of inhalation burn in a patient that has been trapped in a burning house.
☐ True ☐ False
- 2. Haemorrhage controlled refers to a patient whose active bleed was controlled by the triage provider by applying direct pressure with a dry dressing.
☐ True ☐ False
- 3. A patient in cardiac arrest is immediately categorised Red.
☐ True ☐ False

Choose the correct answer:

- 4. Examples of high energy transfers include:
(a) Motor vehicle accident at 60 km/h
(b) Fall from a height of ten meters
(c) Pedestrian vehicle accident at 50 km/h
(d) High velocity gunshot wound
(e) All of the above
- 5. A threatened limb may present as:
(a) Pain in the affected limb
(b) A pale, pulseless affected limb
(c) A weak or numb affected limb
(d) The affected limb is cold and has poor perfusion
(e) All of the above

Important to note that if a patient has emergency signs then a TEMS does need to be calculated to categorise them as **RED**. There should be no delay in getting the patient to the resuscitation area. The first set of vitals may be obtained in the resuscitation area or in the ambulance.

Exposed to heat (e.g. trapped in a house fire, chemical burn to face or mouth) may get a facial laceration. A facial laceration sign may have singed facial hair around their nose/mouth and should be categorised as **RED**.
A patient who is not breathing, a patient with a low oxygen saturation (e.g. 88% on 15L or cardiac arrest).

a	acute	Burn – circumferential
b		Burn – chemical
c		Poisoning / Overdose
d		Diabetic – glucose over 11 and ketonuria
e	joint	Vomiting fresh blood
f	around skin)	Pregnancy and abdominal trauma
g		Pregnancy and abdominal pain
h	al	Severe Pain

These include a motor vehicle accident of 60 km/h or a high velocity gunshot wound.

ADULT CLINICAL SIGNS

Learning Objectives:

- Be familiar with the adult emergency signs
- Be familiar with the adult very urgent and urgent signs

3.1 EMERGENCY SIGNS

EMERGENCY
Obstructed airway – not breathing
Seizures - current
Burn – facial /inhalation
Hypoglycaemia – glucose less than 3 mmol/L
Cardiac arrest

It is any NO then takin the resu

Burn - facial inhalation

Any patient in whom the airway has potentially been exposed to hot water burn to face with possible steam inhalation or chemical rapid swelling of the airway. Adults presenting with this emergency signs (eyelashes, eyebrows), carbonaceous material in and a triaged Red.

Other emergency signs for adults include an obstructed airway, convulsing, and hypoglycaemia with a glucose less than 3 mmol/L.

3.2 VERY URGENT SIGNS

VERY URGENT	
High energy transfer (severe mechanism of injury)	Focal neurology – (stroke)
Shortness of breath - acute	Aggression
Level of consciousness reduced / confused	Threatened limb
Coughing blood	Eye injury
Chest pain	Dislocation of large joint (not finger or toe)
Stabbed neck	Fracture - compound (with a break in skin)
Haemorrhage – uncontrolled (arterial bleed)	Burn over 20% TBSA
Seizure – post ictal	Burn – electrical

High energy transfer (severe mechanism of injury)

In our context this refers to high speed injuries. Examples of this include a motor vehicle accident at 40 km/h or more, pedestrian vehicle accident, a fall from a height of 3m or more.

IS THE CHILD BREATHING?

If active, talking, or crying, the child is obviously breathing. If in any doubt you must **ASSESS** three things to check if the child is breathing (see Figure 6):

- **LOOK** - to see if the chest is moving.
 - **LISTEN** - for any breathing sounds.
 - Are they normal?
 - **FEEL** - Can you feel the breath at the nose or mouth of the child?
- If the child is not breathing (or you are not sure if there is breathing), you need to take the child to the resuscitation area and if the child needs to be artificially supported by ventilating the child with a bag valve mask (BVM) device whilst the child is further assessed and managed appropriately.

IS THE BREATHING OBSTRUCTED?

Noisy breathing can be a sign that the airway maybe threatened or partially obstructed. This is most common in patients with a decreased level of consciousness, upper respiratory tract infections or aspiration of foreign bodies. Obstructed breathing can also be due to blockage by the tongue or the patient's own secretions if these are not being swallowed.

ARE THERE ABNORMAL RESPIRATORY NOISES?

Are there any noises heard when breathing in? A harsh noise on breathing in is called **stridor**, a short noise when breathing out in young infants is called **grunting**. Both noises are signs of severe respiratory problems.

NB: A blocked nose is an extremely common cause of noisy breathing, but it is NOT life-threatening.

DOES THE CHILD SHOW CENTRAL CYANOSIS?

DEFINITION: Cyanosis occurs when there is an abnormally low level of oxygen in the blood. This produces a bluish or purplish discoloration of the tongue, the inside of the mouth and the skin. This sign may be absent in a child who has severe anaemia.

To **ASSESS** for central cyanosis:

LOOK - at the mouth and tongue. A bluish or purplish discoloration of the tongue and the inside of the mouth indicates central cyanosis.

MEASURE - If oxygen saturation monitoring is available you can check the child's oxygen levels. First ensure that the probe is correctly sited and that a good regular trace is showing on the monitor. If oxygen saturation is < 92% in room air the child has low oxygen levels and this is an emergency.

DOES THE CHILD HAVE SEVERE RESPIRATORY DISTRESS?

To **ASSESS** whether the child has severe respiratory distress check for the following signs:

- Is the child having trouble getting breath so that it is difficult to talk, eat or breathe?
- Is the child breathing very fast?
- Does the child have severe chest indrawing? This can be intercostal (between the ribs), subcostal (below the ribcage), suprasternal (above the sternum) or sternal indrawing (the breastbone is sucked in on inspiration).
- Does the child have nasal flaring or a grunting noise on expiration?
- Is the child using the accessory muscles of the neck for breathing? This can cause the head to nod or bob with every breath. This is particularly seen in young babies.
- Exhaustion: If the child's breathing is very laboured, especially if it has been like this for some time, then s/he may become exhausted? If this happens then the signs of increased work of breathing can actually decrease as the child is tired - this is a very dangerous sign.

If you see these signs then it is likely that the child has severe respiratory distress.



Figure 6: Look, listen and feel

signs
gms

injury. These emergency signs relate to
tion. Other and are easily remembered
rich, when identified, should alert you to
h. It is important to note that if a patient
calculated. There should be no delay
tials may be taken in the resuscitation

oed

than 92%

following:

or more

sebleed)

) OR Unresponsive (U)

st-ictal and not alert

ore of the following:

ec or more

ny time - glucose less than 3 mmol/L

as if the child has airway or breathing
is you need to know:
e child breathing?
e airway obstructed?
e child blue (centrally cyanosed)?

ing. An airway
attention before you move on to
of the alphabet represent the two
there is no problem with the airway or
y C.

PAEDIATRIC CLINICAL SIGNS

Learning Objectives:

- Be familiar with the paediatric emergency signs
- Understand the ABC-c-c-DO approach for emergency signs
- Be familiar with the paediatric very urgent and urgent signs

4.1 EMERGENCY SIGNS: The ABC-c-c-DO approach

Triage of patients involves looking for signs of serious illness or the Airway/Breathing/Circulation/Coma/Convulsion-Dehydration as "ABC-c-c-DO". Each letter refers to an emergency sign which a patient who is seriously ill and needs immediate intervention has any emergency signs then a TEWS does NOT need to be taken in taking the patient to the resuscitation area. The first set of V area or in the ambulance on the way to the hospital.

EMERGENCY	
<u>A</u> irway and <u>B</u> reathing	Not breathing or reported apnoea Obstructed breathing Central cyanosis or SpO ₂ less than 92% Respiratory distress (severe)
<u>C</u> irculation	Cold hands + 2 or more of the (i) pulse weak and fast (ii) capillary refill time > 3 sec (iii) lethargic
<u>C</u> oma	Uncontrolled bleeding (not nose) AVPU: Responds only to Pain (P) Confusion
<u>C</u> onvulsions	Convulsing or immediately post-convulsions
<u>D</u> ehydration	Diarrhoea or vomiting + 2 or more of (i) Lethargy/troopy infant (ii) Very sunken eyes (iii) Skin pinch very slow - > 2 s
<u>O</u> ther	Facial /inhalation burn Hypoglycaemia recorded at admission Purpuric rash

HOW TO Triage?

Keep in mind the ABC-c-c-DO steps:
Airway, Breathing, Circulation, Coma,
Convulsion, Dehydration and Other.

4.1.1 **AB** AIRWAY AND BREATHING ARE USUALLY ASSESSED TOGETHER

The letters **A** and **B** in "ABC-c-c-DO" represent "airway and breathing". It is evident that an open (patent) airway is needed for breathing or breathing problem is life-threatening and must receive you other systems. It is therefore convenient that the first two letters most important areas to look for emergency or priority signs. If breathing, you should look for signs in the areas represented by

To assess breathing problem:

- Is the child breathing?
- Is the sound normal?
- Is the chest moving?

check oxygen saturation levels if you have well sited and that you have a good trace on all, this child has an emergency **sign in severe respiratory distress, oxygen saturation in severe respiratory distress, rather late the emergency sign.**

quickly continue the assessment for them, you should initiate appropriate

IT NEED TO CHECK THE CIRCULATION ANY LESS FOR OTHER SIGNS OF CIRCULATORY LETHARGY.

child has circulation problems you
d have cool or cold hands? **IF YES**
any refill time (CRT) 3 seconds or more?
prolonged is the pulse weak and fast?
lethargic

are central pulse (a pulse nearer to the
to look is at the middle of the upper
look for the femoral pulse in the groin.
ask, decide if it also seems fast. This is a
central pulse is weak and fast, the child



Figure 7: Feeling the brachial pulse in an infant

To assess for coma and convulsions

You need to know:

- (i) Is the child's level of consciousness disturbed?
- (ii) Is the child convulsing now?

ently shaken. A little shake to the arm or
essful, apply a firm squeeze to the nail
use some pain. A child who does not
being shaken or to pain is unconscious.
the consciousness level of a child, a
is used;

alert, but responds to voice, is lethargic.
child may or may not respond to pain. **Any
scale of 'P' or 'U' is an emergency and
to the resuscitation area immediately.**

IS THE CHILD CONVULSING NOW OR POST-ICTAL?

This assessment depends on your observation of the child and on the history from the parent. Children who have a history of convulsion, but are alert during frage, will need a complete clinical history and investigation by a clinician, but at frage they are not assigned an emergency clinical sign, as they do not usually require any resuscitation immediately.

The child may be seen to have a convulsion during the frage process or while waiting in the outpatient department. You can recognize a convulsion by the sudden loss of consciousness associated with uncontrolled jerky movements of the limbs and/or the face. There is stiffening of the child's arms and legs and uncontrolled movements of the limbs. The child may lose control of the bladder, and is unconscious during the convulsion and drowsy afterwards.

Sometimes, in small infants, the jerky limb movements may be absent, but there may be more subtle twitching movements of the face, mouth, eyes, hands or feet. You have to observe the infant carefully.

4.1.4 D SEVERE DEHYDRATION ASSESSMENT

In this section we will look at the assessment of severe dehydration in the child with diarrhoea or vomiting. If the child is severely malnourished the signs of dehydration are not as reliable.

DOES THE CHILD HAVE DIARRHOEA OR VOMITING?

This information comes from the parent or guardian.

If the child has no diarrhoea or vomiting, do not check for dehydration.

If the child has diarrhoea or vomiting assess for severe dehydration.

A history of diarrhoea or vomiting and the presence of two or more of the signs in the box above means the child has severe dehydration and needs to be taken to the resuscitation area immediately.

IS THE CHILD LETHARGIC?

In the older child lethargy is quite easy to assess. You have already assessed the state of consciousness of the child using the AVPU scale. Now observe if the child appears drowsy and does not show interest in what is happening around him/her. A lethargic child may not look at the mother or watch your face when you talk. The child may stare blankly and appear not to notice what is going on around him/her.

Does the child know his/her name and answer questions sensibly? If the child responds to voice but remains drowsy, he/she is lethargic. In the younger child, signs of lethargy are harder to assess.

DOES THE CHILD HAVE VERY SUNKEN EYES?

Look at the child's eyes to determine if they appear unusually sunken in their sockets (see Figure 8). Ask the mother if the child's eyes are more sunken than usual.

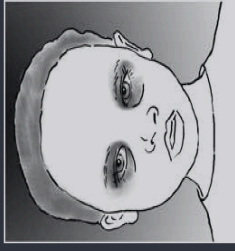


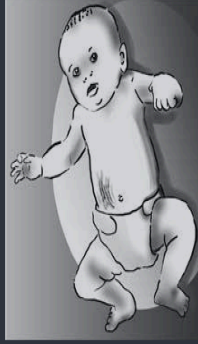
Figure 8: Sunken eyes

DOES A SKIN PINCH GO BACK VERY SLOWLY (2 SECONDS OR MORE)?

This is a simple test to look at how elastic the skin is. If the child is not dehydrated, the skin will be elastic and, when pinched and released, will return to normal straight away. Try this on yourself.



Figure 9: Skin pinch



MEASURE: In any child with respiratory distress, you should check an oxygen saturation monitor. First ensure that the probe is in the correct position on the monitor – if the oxygen saturation is less than 92% in room air, the child needs oxygen therapy. **However if a child is obviously in respiratory distress, saturation levels do not need to be checked at triage to confirm the need for resuscitation area immediately.**

child to the resuscitation area immediately.

If the child is breathing adequately, go to the next section to emergency signs. If the child has an airway or breathing problem and then quickly resume the assessment.

4.1.2 C CIRCULATION ASSESSMENT

FIRST FEEL THE CHILD'S HANDS – IF THEY ARE WARM YOU CAN PROCEED. IF THEY ARE COLD, HOLD THEM FOR A MINUTE OR TWO AND TRY AGAIN. IF THEY ARE STILL COLD, MOVE ON TO THE NEXT EMERGENCY SIGN.

IF THE HANDS ARE COLD OR COOL – YOU NEED TO RAPIDLY ASSESS THE PATIENT FOR SHOCK: PULSE VOLUME AND RATE; CAPILLARY REFILL TIME AND

IS THE PULSE WEAK AND FAST?

The radial pulse (the pulse at the wrist) should be felt: if this is strong and not obviously fast, the pulse is adequate; no further assessment is needed. The radial pulse is used as an initial screen because it is easy to access without undressing the patient.

To assess if the need to know:

- Does the child
 - Is the capillary
 - If the CRT is
 - Is the child

If the radial pulse is difficult to find, you need to look for a more proximal pulse. The brachial pulse is the next best place to feel for a pulse in an infant (less than one year of age). The brachial pulse is the pulse in the arm. The brachial pulse, if the child is lying down, you could feel in the upper arm. The brachial pulse, if the child is sitting up, you could feel in the upper arm. The brachial pulse, if the child is standing, you could feel in the upper arm. The brachial pulse, if the child is lying down, you could feel in the upper arm. The brachial pulse, if the child is sitting up, you could feel in the upper arm. The brachial pulse, if the child is standing, you could feel in the upper arm.

Note that we do not recommend blood pressure to assess for shock at triage because of two reasons:

1) Low blood pressure is a late sign of shock in children and they will already have other obvious signs and 2) the blood pressure in children is less predictive at triage than in adults.

4.1.3 C-C COMA AND CONVULSIONS ASSESSMENT

IS THE CHILD IN A COMA?

A child who is awake is obviously conscious and you can move to the next component of the assessment. If the child is asleep, ask the mother if the child is just sleeping. If there is any doubt, you need to assess the level of consciousness.

Try to wake the child by talking to him/her, e.g. call his/her name loudly. A child who does not respond to this should be gently shaken. If the child does not wake, the caregiver's leg should be enough to wake a sleeping child. If this is unsuccessful, the caregiver should call the child's name loudly.

A Is the child Alerts if not:

V Is the child responding to Voice?
If not,

P Is the child responding to Pain?

U The child who is **Un**responsive to voice (or being shaken) **AND** to pain is **Unconscious**.

To help you assess
simple scale (AVPL)

A child who is not
An unconscious ch
child with a coma
needs to be taken

Tiny baby (less than two months of age)

If the child appears very young, ask the mother his age. If the child is obviously not a young baby, you do not need to ask this question. Small babies are more difficult to assess properly, more prone to getting infections (from other patients), and more likely to deteriorate quickly, if unwell. All tiny babies of under two months of age should therefore be seen very urgently.



ADDITIONAL TASK
Tiny baby →
Refer to SHCP

Inconsolable crying/ severe pain

The inconsolable crying child is conscious but cries constantly and will not settle. Ask the parent of caregiver if the child is upset/frightened by the unfamiliar environment or whether this is the presenting complaint. If a child has severe pain and is in agony, s/he should be prioritized to receive very urgent assessment and pain relief. Severe pain may be due to severe conditions such as acute abdomen, meningitis, etc. You are not required to do a formal pain scale assessment, but for your information there are some examples of pain scales appended at the end of part four.

Presenting complaint - 'more sleepy than normal'



ADDITIONAL INVESTIGATION
Reduced level of consciousness
→ do finger prick glucose test



WARNING
If glucose less than 3 mmol/L
→ take to resuscitation area

When the mother complains that her child that is more sleepy than normal (NOT just in natural sleep), the child may be in the early stages of serious conditions such as meningitis, hypoglycaemia, septicemia etc, and will need to be identified and managed very urgently. Even if the child appears alert at triage still assign a Very Urgent clinical sign as we have to take seriously the mothers concern.

Poisoning or Overdose

A child with a history of swallowing drugs or other dangerous substances needs to be assessed very urgently, as s/he can deteriorate rapidly and might need specific treatment depending on the substance taken. The mother will tell you if she has brought the child because of possible intoxication. Consult SHCP for advice regarding very urgent management.



ADDITIONAL TASK
Poisoning/Overdose
→ Refer to SHCP

Focal neurology acute

A child who is fully conscious but has new focal neurological signs will need very urgent assessment. In this case, the child does not require emergency treatment because they do not have any ABCD emergency signs, but may need urgent imaging and intervention. Examples of focal neurological signs include cranial nerve palsies and acutely paralysed limbs - their presence may indicate a form of vascular stroke and management may be time dependent - the child may have a weakness on one side of the body and may not be able to move as normal. The parent may also complain that their child has a new onset of a squint or a paralysed face.

Severe Trauma – severe mechanism of injury

Usually this is an obvious case, but one needs to think of motor vehicle accident injuries, falls from a height greater than one metre in height, gunshot wounds and other traumatic mechanisms of injury in this category. Remember the ABC-c-c-DO assessment would already have identified any life-threatening problems requiring immediate resuscitation.



ADDITIONAL TASK
Bleeding → apply pressure to the site of trauma and cover open wounds
Pain → check with SHCP for analgesia initiation

Burns

These include: Circumferential, Electrical, Chemical and any burn involving 10% or more of body surface area.

Burns are extremely painful and children who seem quite well can deteriorate rapidly.



WARNING
Any major burn → take to resuscitation area



ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation

ur fingertips, as this is painful. Pinch the child's body, you should pick up all the second and then release. See whether

a plus any two of the following signs:

- Perform a finger prick glucoatest in the following cases:
- Reduced level of consciousness
 - Unable to sit or move as usual
 - Current or recent seizure
 - Known with diabetes
 - Severe malnutrition

t to be seen immediately. They should tion of less than 3 mmol/L.

essed to see if the spots blanch with are is applied). If the spots are non- then there is a risk of meningococcal and. If you are unsure about a

ency signs in paediatric patients. The actively. It is important to note that a d to be calculated. There should be the first set of vitals may be taken in the

uld NOT be calculated at triage, the the resuscitation area. e and should be seen immediately. undered over directly to a SHCP.

the child has any VERY URGENT signs.

VERY URGENT
Tiny baby - younger than 2 months
Inconsolable crying / severe pain
Presenting complaint - More sleepy than normal
Poisoning or overdose
Focal neurology acute
Severe mechanism of injury
Burns (circumferential, electrical, chemical, 10% or more)
Eye Injury
Fracture – open or threatened limb
Dislocation of larger joint (not finger or toe)

the umbilicus and the side of the abdomen. Avoiding using your skin in a vertical (head to foot) direction and not across the creases of the skin and the fat tissue underneath. Pinch for one second, then release. The skin should return to its normal colour. If the skin goes back very slowly (2 seconds or more).

Severe dehydration is present if the child has a history of diarrhoea or vomiting, sunken eyes or very slow skin pinch (2 seconds or more).

4.1.5 ○ OTHER EMERGENCY SIGNS

Facial Inhalational burns

Any child in whom the airway has potentially been exposed to heat (e.g. trapped in a house fire, hot water burn to face with possible steam inhalation or chemical burn to face or mouth) may get rapid swelling of the airway. Children presenting with this emergency sign should be triaged Red.

Glucose <3mmol/L at any time

Children with a low blood sugar are often very sick and need to be triaged Red if found to have a blood glucose concentration <3mmol/L.

Purpuric Rash

Children who present with a complaint of a rash - should be a pressure from fingertip or glass test (i.e. turns white when pressure is applied) - blanching (i.e. does not turn white when pressure is applied) - disease. Children presenting with this emergency sign should be triaged Red. Children presenting with a rash ask the senior healthcare professional for advice.

You have now learned how to recognize the obvious emergency signs. The next section will cover the very urgent and urgent signs respectively. If a child has any emergency signs then a TEWS does NOT need to be taken. There should be no delay in taking the patient to the resuscitation area. The triage area is the resuscitation area or in the ambulance on the way to the hospital.

- If any **EMERGENCY** signs have been found the TEWS should not be taken. The child is within the **RED** category and should be taken to the resuscitation area.
- Children within the **RED** category need emergency care.
- Always ensure that the child with emergency signs is hospitalized.
- If there are no **EMERGENCY** signs, check to see whether the child is within the **ORANGE** or **YELLOW** category.

4.2 VERY URGENT SIGNS

If the child does not have any of the emergency ABC-C-C-DO signs, the triage provider proceeds to assess the child on the very urgent signs. This should not take more than a few seconds. Some of these signs will have been noticed during the ABCD triage discussed so far, and others need to be re-checked. Follow the list of very urgent signs to quickly complete this section. Decide whether the child has any very urgent or urgent signs and need prompt management.

The presence of obvious very urgent signs does not automatically make the child's triage priority **ORANGE**. Always calculate the TEWS and check key additional investigations to ensure that the child does not need to be assigned to the **RED** category and taken for emergency care.

The frequency with which children showing these very urgent signs appear in your emergency centre depends on the local epidemiology.

APPENDIX B: Examples of different pain scales**Behavioral Observation Pain Rating Scale**

Categories		Scoring	
	0	1	2
Face	No particular expression or smile; disinterested	Occasional grimace or frown, withdrawn	Frequent to constant frown, clenched jaw, quivering chin
Legs	No position or relaxed	Uneasy, restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid or jerking
Cry	No crying (awake or asleep)	Moans or whimpers, occasional complaint	Crying steadily, screams or sobs, frequent complaints
Consolability	Content, relaxed	Reassured by occasional touching, hugging, or talking to, Distractable	Difficult to console or comfort

Each of the five categories: **(P) Face; (L) Legs; (A) Activity; (C) Cry; (C) Consolability** is scored from 0 – 2, which results in a total score between 0 and 10.

Observe the child and score the child's pain according to the 'FLACC' scale.

'Faces' Pain Rating Scale**4.3 URGENT SIGNS**

These children do not have any of the emergency ABC-c-c-DO signs nor do they have any of the very urgent signs. The triage provider should proceed to assess the child on the urgent signs. This should not take more than a few seconds.

Even if there are obvious urgent signs – calculate the TEWS and carry out additional investigations to ensure that the patient is not possibly within the **RED** or **ORANGE** category.

Some respiratory distress

When you assessed the airway and breathing, did you observe any respiratory distress? If the child has severe respiratory distress, this is an emergency. There may however, be signs present that you do not think are severe, e.g. some lower chest wall indrawing (but not severe), or slight increase

ADDITIONAL INVESTIGATION
RR scoring 1 point or more → measure oxygen saturation

URGENT

Some respiratory distress

Some Dehydration - Diarrhoea or diarrhoea and vomiting + 1 or more of the following:

- (i) sunken eyes
- (ii) restless / irritable
- (iii) dry mouth, increased urine output
- (iv) dry mouth, no tears
- (v) crying without tears
- (vi) skin pinch slow-less than 2 sec

Some Dehydration - Unable to drink/feed OR vomits even with more of signs (i) – (vi) above

Malnutrition (visible severe wasting)

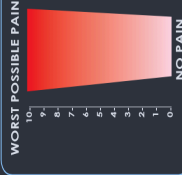
Malnutrition Oedema (pitting oedema of BOTH feet)

Unwell child with known diabetes

Any other burn less than 10%

Closed fracture

Dislocation of finger or toe



eds to be determined using the Rule of 20. For any child with a major burn (≥ 20%) analgesia should be commenced. Follow the provincial burns management protocol (e.g. Severe Burns 2011) including analgesia.

one-year-old child. Refer to Table in the Management of Severe Burns manual for

ADDITIONAL TASK
If occurred recently (within 3hrs) → turn child onto their side to cool the burnt area with water. For example, by running cool tap water over for 30 minutes. The child should then be placed in a clean sheet or blanket to avoid further heat loss. If the child is in a sling wrap if dry sheet or towel will also be suitable.



ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation

ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation



ADDITIONAL TASK
Dislocation of large joint → Hand over to SHCP. Pain → check with SHCP for analgesia initiation

ould be calculated and key that the child does not need to be y care.

within 10 minutes.

the health worker in the Orange area,

ther the child has any URGENT signs.

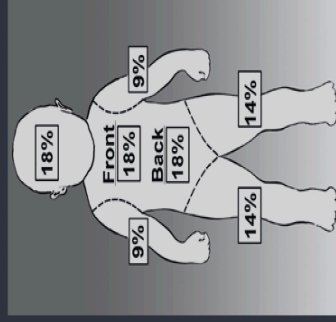


Figure 10: Rule of Nines for burn surface area estimation in a one year old child

Fracture – open

An open fracture may be associated with a large volume of blood loss that may be very obvious (external) or may be concealed (internal) – perhaps only recognisable by swelling around the fracture area. This needs very urgent attention.

Threatened limb

A threatened limb presents as:

- P Pain
- P Pallor
- P Pulselessness
- P Paralysis or pins & needles
- P Capillary Refill Delay
- P Temperature

Dislocation of large joint (not finger or toe)

Apart from being very painful, a large joint dislocation may compromise blood and nerve supply to the limb distal to the affected joint. Damage to the neurovascular bundle implies serious injury and needs very urgent attention.

- If any **VERY URGENT** signs have been found the TEWS should be checked to ensure additional investigations should be checked to ensure assigned to the **RED** category and taken for emergency
- Children within the **ORANGE** category should be seen
- Handover all **ORANGE** category children personally to
- Always check for additional tasks that should be done
- If there are no **VERY URGENT** signs, check to see whether



ADDITION
if the burn is still wet, it is still wet, water, for over the burnt area to be dried and wrapped to prevent hypothermia. The burn should be covered with a clear, available, or a clear



ADDITION
Pain → ch for analge

WARNING
If oxygen saturation below 92% → give oxygen and move to resuscitation area



or more signs of dehydration

enough fluid to show one or more of

mouth

g without tears

g with slow - less than 2 seconds

difficult to quantify, they need to avoid becoming severely dehydrated or

ADDITIONAL TASK
Diarrhoe → start oral rehydration therapy (ORT)



vomiting everything may have serious implications. Examples include septicaemia, meningitis, heart conditions, acute surgical illness, pneumonia etc.

ADDITIONAL TASK
Vomiting only and dehydration → consult with SHCP



WARNING
If glucose is below 3 mmol/L → move to resuscitation



to pick up because their muscles are very tired and their upper thighs, buttocks and arms may be wasted. They may also be dehydrated and look rapidly ill. These body areas should be examined for signs of dehydration, glycaemia and infections and need to be managed according to the paediatric resuscitation guidelines to prevent complications.

ADDITIONAL INVESTIGATION
Malnutrition with visible severe wasting → do a finger prick glucose test and check with SHCP

WARNING
If glucose is less than 3 mmol/L → move to resuscitation



Unwell child with known diabetes

Children with diabetes often lose control of their sugar levels when they are unwell from acute illness and infections e.g. acute respiratory infections, (upper or lower) urinary tract infections or diarrhoea.

They often develop high or low blood glucose levels because their insulin needs may increase or decrease whilst they are unwell and they are not eating as usual because of loss of appetite, nausea or vomiting.

Burn less than 10% of body surface area

The size of the burn needs to be determined using the palmar method (patient's outstretched open palm including the fingers is ≈ to 1% of the body surface area).

ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation



ADDITIONAL TASK
Burn less than 10% → follow the provincial Burns management guidelines. The burn is small and the child is less than 3 years old. It is still worthwhile to cool the burnt area with water, for example, with cool tap water for at least 30 minutes. The burn should then be dried and covered with cling wrap or a clean dry sheet. The child should also be wrapped in a blanket and kept warm



Closed fracture (no break in the skin)

These children often present with pain on moving or refusal to move the affected limb, there may be deformity of the limb evident.

ADDITIONAL TASK
Closed fracture → if required get a wheelchair or stretcher and immobilise the affected limb with a simple splint e.g. a padded wire splint or triangular bandage that does not interfere with the blood supply or nerve bundle, and provide analgesia according your site's guidelines.



Dislocation of finger or toe

This often causes an obvious deformity, the dislocated digit will need to be reduced under some form of anaesthesia.



ADDITIONAL INVESTIGATION
History of diabetes → do a finger prick glucose test



WARNING
If glucose is below 3 mmol/L → move to resuscitation

WARNING
If glucose result is 'HI' → check with SHCP

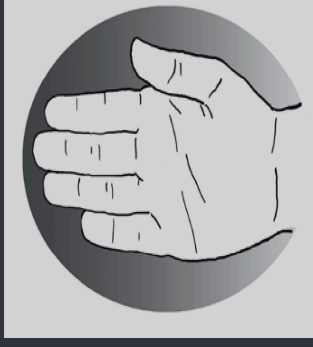


Figure 11: Palmer method illustrating 1% of the body



ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation



ADDITIONAL TASK
Pain → check with SHCP for analgesia initiation

in breathing rate. In this case, the child does not require emergency treatment but will need urgent assessment. Understanding the level of severity of respiratory distress come with practice and experience. If you have any doubts, consult a senior health care professional immediately.

Some Dehydration - Diarrhoea or diarrhoea and vomiting + 1

Infants and children with diarrhoea or vomiting may have lost the following signs of dehydration:

- | | |
|--------------------------------------|-------------------|
| (i) sunken eyes; ask the mother | (iv) dry nappies |
| (ii) restless/irritable | (v) crying |
| (iii) thirst/ decreased urine output | (vi) skin pinches |

Because these children have losses that are ongoing and difficult to replace, commence oral rehydration therapy as soon as possible to a shocked i.e. signs that would now place them into the RED category. Most of them are thirsty and their mothers need to offer them extra fluids according to the Oral Rehydration Corner guidelines. Infants and young children that are breast-fed should be encouraged to continue to do so whilst awaiting further urgent assessment.

Unable to drink/feed OR vomits everything

Infants and children that are not well enough to feed or are vomiting or having diarrhoea may have lost the following signs of medical or surgical conditions affecting any of the body systems:

Vomiting only and dehydration:

- beware the child may have an acute surgical bowel problem
- oral fluids may be contraindicated
- seek advice from a senior health professional

Malnutrition with visible severe wasting



ADDITIONAL INVESTIGATION
Malnutrition with visible severe wasting - do a fingerstick glucose test and check with SHCP

Severe wasting is a form of severe malnutrition. These children are easy to waste and they have loose skin folds particularly noticeable on the upper arms. If you are concerned that a child might be malnourished, severely malnourished children are prone to hypothermia, hypoglycaemia and need to be managed urgently according to the WHO severe malnutrition guidelines.

Malnutrition with pitting oedema of both feet

This is another form of severe malnutrition. It is as dangerous as the one described above and is recognised by the pitting oedema of both feet; they may also present with general body swelling. If you are concerned that a child might be malnourished look rapidly at the child's feet and check for oedema of both feet. These children often appear listless and apathetic and have skin and hair changes. They are also prone to hypothermia, hypoglycaemia and infections and need to be managed urgently according to the WHO severe malnutrition guidelines to prevent complications.



Learning Objectives:

- Be familiar with the different age appropriate versions of the TEWS
- Understand how to calculate the TEWS in paediatric and adult patients

The TEWS is a composite score representing physiologic parameters at triage. There are different age appropriate versions: the **younger child TEWS** is for patients smaller than 95cm or younger than 3 years (see Figure 12); the **older child TEWS** is for patients 95cm to 150cm or 3 years to older than 12 years (see Figure 13); and the **adult TEWS** is for patients older than 12 years or taller than 150cm. Older children (see Figure 13), and adults, who are not to be reassessed should be reassessed, if they are taller 150cm, then the adult version should be used. This standardised scoring system has been validated and many of the boxes in the TEWS calculator are shaded grey. These boxes cannot be assigned a score. This means that for temperature, for example, it is only possible to score 0 or 2 points, depending on the value of the recording.

YOUNGER CHILD TEWS						
YOUNGER THAN 3 YEARS / SMALLER THAN 95 cm	3	2	1	0	1	2
Mobility				Normal for age		Unable to move as normal
RR	less than 20	20 - 25		26 - 39		40 - 49
HR	less than 70	70 - 79		80 - 130		131 - 159
Temp		Feels Cold Under 35°		35° - 38.4°		Feels Hot Over 38.4°
AVPU				Alert	Reacts to Voice	Reacts to Pain
Trauma				No	Yes	Unresponsive

Figure 12: Younger Child TEWS (younger than 3 years)

OLDER CHILD TEWS						
3 to 12 YEARS OLD / 95 to 150 cm tall	3	2	1	0	1	2
Mobility				Normal for age		Unable to walk as normal
RR	less than 15	15 - 16		17 - 21	22 - 26	27 or more
HR	less than 60	60 - 79		80 - 99	100 - 129	130 or more
Temp		Feels Cold Under 35°		35° - 38.4°		Feels Hot Over 38.4°
AVPU		Confused		Alert	Reacts to Voice	Reacts to Pain
Trauma				No	Yes	Unresponsive

Figure 13: Older Child TEWS (age 3 - 12 years)

- If any **URGENT** signs have been found the **TEWS** should investigations checked to ensure that the child does not have **RED** or **ORANGE** category and taken for emergency or specialist care.
- Children within the **Yellow** category should be seen within 24 hours.
- Always check for additional tasks that should be done if there are no **URGENT** signs, calculate the **TEWS** and check if there are any **URGENT** signs.
- If there are no **URGENT** signs, calculate the **TEWS** and check if there are any **URGENT** signs.

Assessment Questions

Clearly indicate whether the following statements are true or false.

1. Grunting and nasal flaring are signs of severe respiratory distress. ☐ True ☐ False
2. Oxygen saturation levels should always be checked in children with respiratory distress. ☐ True ☐ False
3. If no emergency sign is identified in step one, but an urgent sign is identified, the patient is immediately taken to the majors area for urgent assessment. ☐ True ☐ False

Choose the correct answer:

4. In the ABC-c-c-DO approach ABC stands for airway, breathing, circulation, disability, oxygenation.
What does c-c-D stand for?
(a) convulsions, chest pain, dehydration
(b) coma, cancer, disabilities
(c) coma, convulsions, dehydration
(d) coma, craniopharyngioma, dehydration
(e) chronic pain, constipation, dehydration
5. The following are emergency signs
(a) oxygen saturation levels more than 92%
(b) facial or inhalation burn
(c) stridor, snoring and secretions
(d) closed fracture
(e) b and c above

TEWS				
	1	2	3	
g	With Help	Stretcher/ Immobile		
	15 - 20	21 - 29		more than 29
0	101 - 110	111 - 129		more than 129
9		more than 199		
4*		Hot OR Over 38.4°		
	Reacts to Voice	Reacts to Pain	Unres- ponsive	
	Yes			

DO NOT UPSET THE CHILD
this affects your observations for TEWS

paths for 30 seconds and then multiplying
child is quiet. If the child is small it is best
o count the respiratory rate as doing this
ner than touching the child.

count the heart rate by feeling the pulse
art rate monitor can be used (ideally
nitors can be unreliable and can

ercury thermometer (preferably a low-
ken routinely.

art then the triage provider needs to
the child responds to voice and is then

is the patient is not fully alert and
ENDS ONLY TO YOUR VOICE. Another way
cribing this = 'lelmarig'

is the patient is not alert
oes not respond to your voice but
ENDS ONLY TO A PAINFUL STIMULUS

MOBILITY

Mobility is observed by noting the mode in which the patient has to be mobilised. A small baby is carried by the carer – as might a small infant or child. You need to assess if the level of movement is normal for that particular child. In the event that the child or adult is in a wheelchair due to permanent paralysis 2 points are given for mobility as the child in a wheelchair is more at risk.

TRAUMA

Trauma is present if there is ANY injury to the patient within the past 48 hours.

5.2 Terminology and key concepts

- Confusion:** An older child may be reported to have confused behaviour or this may be discovered at triage. A confused child appears disorientated, s/he may not be interacting normally with the carer or the environment, s/he may be delirious or hallucinating. A confused child may be talking a lot and aggressive or s/he may be quiet and fearful looking. This sign is difficult to detect in younger children who are usually pre-verbal - so it is only included on the TEWS for the older child.
- Acute vs Chronic:** Many patients in South Africa are chronically unwell with serious conditions such as TB or HIV/AIDS. In these patients, daily resting physiology such as respiratory rate may be abnormal. This is important to bear in mind, as their TEWS will be high and they will be given an overly high triage code (they may be overtriaged). It is not for the inexperienced triage provider to decide whether this is the case or not; as sick patients may be missed. However, it is appropriate to ask a senior doctor or sister to review the patient's triage code.
- The younger and older child:** Younger and older children have different resting vital signs to adults. These are often difficult to obtain, and the patients often cry, which pushes up their respiratory and heart rate. The values still need to be recorded as you measure them, so that a correct score can be given in the TEWS, but tell the senior doctor or sister that the child was crying when they were recorded. These staff members can then decide whether the child has been overtriaged or not.

Assessment Questions

Clearly indicate whether the following statements are true or false:

- If a child is nine years old we use the older child TEWS.
☐ True ☐ False
- Always undress a child and weigh them so that it is easier to obtain their vital signs.
☐ True ☐ False
- To accurately obtain a respiratory rate always start when the patient is at rest, count respirations for 30 seconds and multiply by two.
☐ True ☐ False

Choose the correct answer:

- The adult TEWS consists of the following parameters:
(a) Mobility, respiratory rate, heart rate, temperature and blood pressure
(b) Mobility, capillary refill time, heart rate, temperature and blood pressure
(c) Mobility, respiratory rate, heart rate, temperature, systolic blood pressure, AVPU and trauma
(d) Mobility, respiratory rate, heart rate, temperature, AVPU and trauma
(e) Mobility, respiratory rate, oxygen saturation level and blood glucose concentration
- The older child TEWS consists of the following parameters:
(a) Mobility, respiratory rate, heart rate, temperature and blood pressure
(b) Mobility, capillary refill time, heart rate, temperature and blood pressure
(c) Mobility, respiratory rate, heart rate, temperature and trauma
(d) Mobility, respiratory rate, heart rate, temperature, AVPU and trauma
(e) Mobility, respiratory rate, oxygen saturation level and blood glucose concentration

OLDER THAN 12 YEARS / Taller than 150 cm tall)	ADULT Triage EWS			
	3	2	1	0
Mobility				Walking
RR		less than 9		9 - 14
HR		less than 41	41 - 50	51 - 100
SBP	Less than 71	71 - 80	81 - 100	101 - 199
Temp		Cold OR Under 35°		35° - 38°
AVPU		Confused		Alert
Trauma				No

Figure 14: Adult TEWS (older than 12 years)

5.1 Observations at triage

To calculate the TEWS the first step is to perform the observations required by the TEWS.

RESPIRATORY RATE

The patient's respiratory rate is calculated by counting the breaths by two. This should be done at the first opportunity when the child is lying down. The patient should be lying on their back, not to leave him/her in the mother's lap. Do not undress the child if they are upset. Count the breathing rate by observing the chest rise and fall. Count the number of breaths in 30 seconds and then multiply by two. Alternatively a child may upset him/her. Count the breathing rate by observing the rate of change frequently if the child moves).

HEART RATE

To measure the heart rate, the triage provider can manually count the heart rate for 30 seconds and then multiplying by two. Alternatively a heart rate monitor can be used. A heart rate monitor should be used if the child is not from a saturation monitor – as the heart rate on these monitors can change frequently if the child moves).

TEMPERATURE

The temperature is measured using either an electronic or mercury thermometer (rectal temperature should NOT be taken in children under 5 years of age).

AVPU

AVPU is done by seeing if the child is alert. If not obviously alert, observe the child's response to their (or the parent's) voice. If the child is alert – he is an 'A' – but if he remains drowsy or unresponsive then he is a 'V'. If he makes no response to voice then the triage provider need to assess the response to a painful stimulus (e.g. nailbed pressure or sternal rub) – if the child responds to this s/he is a 'P'. If there is no response to either verbal or pain stimuli the patient is labelled as unresponsive.

V

mean
RESPONSE
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RESPON

P

DIABETES AND HYPERGLYCAEMIA (GLUCOTEST 11 OR MORE)

All patients with a glucose concentration of 11 mmol/L or more require a urine dipstick to check for ketones.

CHILD HAS MALNUTRITION WITH SEVERE VISIBLE WASTING or WITH PITTING OEDEMA OF BOTH FEET

This child is at risk of hypoglycaemia, as well as hypothermia



**ADDITIONAL
INVESTIGATION**
Perform a urine dipstick to
check for ketones



ADDITIONAL INVESTIGATION
Perform a finger prick glucotest
Immediately to exclude hypoglycaemia



WARNING
If glucose is below 3 mmol/L
→ move to resuscitation and
hand over to SHCP



WARNING
For children if oxygen saturation
is below 92% on room air →
Move to resuscitation area and
administer nasal prong or
face mask oxygen

...y such as chest infection requiring



**ADDITIONAL
INVESTIGATION**
Perform a finger prick
glucotest immediately to
exclude hypoglycaemia



WARNING
If glucose is below 3 mmol/L →
move to resuscitation and
hand over to SHCP

...ood sugar level. Tiny babies and



**ADDITIONAL
INVESTIGATION**
Perform a finger prick
glucotest immediately to
exclude hypoglycaemia



WARNING
If glucose is below 3 mmol/L →
move to resuscitation and
hand over to SHCP

...caemic (usually drowsy or confused) or
need a glucotest done at presentation.



WARNING
If glucose is below 3 mmol/L →
move to resuscitation and
hand over to SHCP
If glucose is 11 → hand over to SHCP

ABDOMINAL PAIN OR BACKACHE IN FEMALES

The adult female with abdominal pain may have an ectopic pregnancy leading to severe pain and discomfort as time progresses. She may be at risk for a ruptured ectopic or other complications in pregnancy requiring emergency surgery.

These additional investigations may rapidly identify very ill patients and change their category. They also prevent and/or identify serious complications.



**ADDITIONAL
INVESTIGATION**
Perform a urine dipstick
and urine pregnancy test

Assessment Questions

Clearly indicate whether the following statements are true or false:

1. A patient with a glucose of 11 mmol/L needs to have a urine dipstick done to check for ketones in the urine.
True ☐ False ☐
 2. Check the finger prick haemoglobin on all patients that have a history of diabetes.
True ☐ False ☐
 3. Do a urine dipstick and urine pregnancy test on all adult females presenting with abdominal pain.
True ☐ False ☐
- Choose the correct answer:
4. Perform a finger prick glucotest on the following cases:
(a) Current or recent seizure
(b) Facial burn
(c) Reduced level of consciousness
(d) History of diabetes
(e) a, c and d

5. The following two additional investigations may upgrade a patient to the Red category:
(a) Oxygen saturation levels in children and finger prick haemoglobin
(b) Finger prick haemoglobin levels and urine dipstick
(c) Urine dipstick test and oxygen saturation levels
(d) Finger prick glucotest and oxygen saturation levels
(e) Finger prick glucotest and finger prick haemoglobin

ADDITIONAL INVESTIGATIONS

Learning Objectives:

- Be familiar with the additional investigations and when to use them.
- Know how additional investigations may change the triage.

For **all patients (especially children)** immediate **additional investigations** for potentially serious complications of their presenting condition are listed below and act accordingly.

RESPIRATORY RATE (RR) SCORES MORE THAN 1 POINT ON TEWS:



ADDITIONAL INVESTIGATION
Measure oxygen saturation (for children on finger, toe, hand or earlobe, depending on available saturation probe and co-operation)

A raised RR for age may indicate serious underlying pathology and supplemental oxygen

REDUCED LEVEL OF CONSCIOUSNESS

All patients that are not fully alert (i.e. confused or only responding to verbal or painful stimulus) need to have a finger prick glucose done and should be handed over to the senior healthcare professional.

If the child is not alert, or the caregiver volunteers that the child is more sleepy than normal, this may indicate a serious evolving medical condition such as meningitis or if associated with a history of trauma there could be a traumatic brain injury.

Any sick child who has not been feeding well or has been vomiting may become drowsy because of a low blood glucose. Malnourished children are particularly at risk of hypoglycaemia.

UNABLE TO SIT OR MOVE AS NORMAL

If the patient is unable to walk or move as normal or the caregiver reports that the child is lethargic or unable to move as usual, this may also be a sign of serious illness or of a low blood sugar level.

RECENT SEIZURE/FIT

The patient who is actively fitting will have been taken to the resuscitation area. In any child with a history of recent seizure there could be easily identifiable and treatable causes, including hypoglycaemia, pyrexia (febrile seizure) or high blood pressure.

HISTORY OF DIABETES

All patients with diabetes are at risk of either becoming hypoglycaemic or hyperglycaemic with diabetic ketoacidosis (DKA). All therefore



ADDITIONAL INVESTIGATION
Perform a finger prick glucose test immediately to exclude hypoglycaemia

VOMITING WITHOUT DIARRHOEA AND THE CHILD IS DEHYDRATED

Vomiting alone which is severe enough to result in dehydration may indicate a diagnosis other than simple gastroenteritis e.g. urinary 'tract' infection; a surgical problem with bowel obstruction; diabetic ketoacidosis or even meningitis or another severe infection. It may **not** be appropriate to automatically commence this child on a trial of oral rehydration – and so senior advice should be sought.

IF THE CHILD HAS A CLOSED FRACTURE

These children will be experiencing significant pain both education and pain mobilization of the affected limb will provide some relief from the pain and should be initiated prior to formal assessment

**ADDITIONAL TASK**

Pain → check with SHCP for initiation of analgesia and review
Closed fracture → immobilise affected limb with simple padded splint or a triangular bandage sling and get a wheelchair or stretcher if required



ADDITIONAL TASK
Vomiting without diarrhoea → refer to SHCP for assessment



ADDITIONAL TASK
Chest Pain → perform an ECG to rule out potential AMI and hand over to SHCP

CHEST PAIN

Patients with chest pain may be having an acute myocardial infarct (AMI). An immediate ECG is required to rule out a potential AMI.

ACTIVE ONGOING BLEEDING

Children have a small circulating blood volume, and are at risk of becoming shocked quickly if there is ongoing bleeding from a trauma site

**ADDITIONAL TASK**

Active bleed → apply direct pressure to the site of trauma with a dry dressing and perform finger prick haemoglobin to obtain a baseline and hand over to SHCP

HISTORY OF RECENT BLEEDING - EITHER RECTAL, ORAL OR FROM A SITE OF TRAUMA

This patient may be anaemic from blood loss

**ADDITIONAL TASK**

History of bleeding → check finger prick haemoglobin. If less than 8 g/dl then hand over to SHCP

PRESENTING COMPLAINT IS ABDOMINAL PAIN

Abdominal pain is a common complaint in children and it has wide variety of possible causes including urinary tract infection, diabetic ketoacidosis, hepatitis or other causes. A urine dipstick for children and adults will assist in excluding or diagnosing these.

These additional tasks may assist to rapidly identify very ill patients and prevent and identify serious complications. They also improve quality of care by providing relief from pain.



ADDITIONAL TASK
For a younger child → Place a urine bag, if older child, give container

A senior healthcare professional should be alerted in the following cases:

- Tiny baby younger than 2 months old
- Reduced level of consciousness
- A child in pain including fractures and burns
- Poison ingestion or overdose

Additional Tasks at triage

ADULT PATIENTS		IMMEDIATE TASKS
1. Temperature 38.5° or more	PROBLEM	Paracetamol 1 g orally stat (document in the notes)
2. Temperature 35° or less		Warm the patient with blankets if available
3. Diabetes and hyperglycaemia (glucose test 11 mmol/L or more)		Urine dipstick to check for ketones
4. History of bleeding		Finger prick haemoglobin
5. Bleeding PR, PO or from the site of trauma		Finger prick haemoglobin
6. Abdominal pain or backache in males		Urine dipsticks
7. PV bleeding		Urine dipsticks, Urine pregnancy test Finger prick haemoglobin

ients (especially children) whether triaged **ORANGE, YELLOW OR GREEN** immediate tasks may be indicated to stabilise the child and/or identify or prevent potentially serious conditions. Check if the child has any of the listed below and act accordingly.



ADDITIONAL TASK
Tiny baby under 2 months → refer to SHCP



ADDITIONAL TASK
Poisoning / overdose → refer to SHCP



ADDITIONAL TASK
Pain or inconsolable crying → check with SHCP for initiation of analgesia and review



ADDITIONAL TASK
Pain → check with SHCP for initiation of analgesia and review to the extent of tissue damage.

worthwhile to cool the burnt area with the burnt area for 30 minutes. The child if or blanket to avoid hypothermia. dry sheet or towel will also be suitable.

ADDITIONAL TASK

High temperature → remove excessive clothing and check with SHCP for initiation of analgesia and review



ADDITIONAL TASK
Low temperature → warm the patient with additional blankets for children with a SHCP cap if available and hand over to SHCP



ADDITIONAL TASK
Diarrhoea & vomiting → take child to ORT corner and advise caregiver to start ORT by cup and spoon

ADDITIONAL TASKS

Learning Objectives:

- Be familiar with the additional tasks
- Know when to perform additional tasks

For all patients, **RED, ORANGE, and additional** patient assessment and complication condition.

TINY BABY UNDER 2 MONTHS

Small babies under 2 months are more difficult to assess, their symptoms are often non-specific, they have lower immunity than other children so are more prone to infections, and they deteriorate more quickly. They therefore need to be assessed as a priority.

POISONING OR OVERDOSE

Toddlers are inquisitive and like to explore their environments. They may accidentally ingest a variety of household substances, pesticides or medications. A child who has ingested a poison can deteriorate quickly. They may require a specific antidote and if a poison or medication has been ingested recently immediate intervention may be needed (e.g. activated charcoal). It is therefore important to consult a SHCP for children and adults even if the patient appears stable.

IF CHILD APPEARS TO BE IN PAIN or IS INCONSOLABLY CRYING

This may be due to a medical cause like an ear infection or from severe headache due to meningitis. Or there may be an obvious cause such as a fracture or laceration. Pain is obvious to a carer. It is good practice to initiate analgesia as soon as possible.

BURN

The patient will be experiencing pain and especially children have the potential to deteriorate rapidly from significant fluid losses and may develop hypothermia.

If the burn is recent (<3hrs) immediate intervention may limit its

ADDITIONAL TASK

If the burn occurred recently (within 3hrs) → It is still possible to cool the burn with running cool tap water over the burn for 10-20 minutes. The burn should then be dried and wrapped in a clean sheet or cling wrap if available, or a clear adhesive dressing.

TEMPERATURE 38.5°C OR MORE

A very high temperature will result in physiological changes that may affect the TEWS, and make the patient feel uncomfortable and in some children may be associated with a febrile seizure.

TEMPERATURE 35.5°C OR LESS

Hypothermia might be a sign of severe sepsis with very small babies, ex-premature babies and severely malnourished children being the most at risk.

DIARRHOEA & VOMITING

Even if the child has no or some dehydration, the child is at risk of becoming dehydrated whilst waiting to be seen and should receive oral rehydration to treat and/or prevent further dehydration.

8.1 Pre-Hospital

The use of triage within a pre-hospital setting is generally well accepted and understood universally. Whilst this need is quite obvious (even critical) during major incidents, it's the daily application of triage principles in the despatch of ambulance resources where it has the most benefit. In these scenarios, triage permits the EMS dispatcher to apply rules based decision making to what is an otherwise impossible choice.

The pre-hospital use of triage in the field varies from region to region, but is generally categorised into four priorities (represented by the colours red, yellow, green and blue). Such triage typically uses instability of vital signs to differentiate high from low priority patients. Discrepancies in triage appear when personnel of differing levels of medical experience and qualifications need to assess patients as there are no clear definitions of 'unstable' physiology. The terms 'stable' and 'unstable' are poorly understood and fail to accurately reflect the patient's clinical condition.

Accurate pre-hospital triage is essential for appropriate call out of secondary resources, accurate notification of receiving hospitals, and quality assessment and audit of the ambulance service. This is particularly pertinent in aeromedical callout requests and use of this specialised resource. For these reasons triage tools based on objective physiological discriminators are essential.

8.2 Patient streaming

Triage assigns the patient to an acuity level, which then dictates the amount of time the patient can wait safely before being seen: Red Immediate, Orange within 10 minutes, Yellow within an hour and Green within four hours. It is the recommended practice to "stream" these patient categories to different areas and/or healthcare providers within the facility. The normal streaming pattern would be Reds to resus, Oranges and Yellows to Majors and Greens to Minors, which would be manned by staff dedicated to these areas. For the most part, streaming patients according to acuity will also stream them according to resource use: full monitoring, access to high-powered drugs and interventions with full team response is necessary to effectively treat the Red patient, while a room with a chair and a single practitioner may be all that is necessary to see and treat the patients in the Green stream. How and where patients are streamed depends on the load, manning and infrastructure of the Emergency Centre or Health Facility.

It is important in any system that prioritises order-to-be-seen by anything other than "first-come-first-served" to have a plan to see the lower priority patients. Streaming is a possible mechanism to achieve this. Without streaming, the Green patient will keep being pushed to the back of the queue by the patient of higher acuity, who by necessity should be seen first. With streaming, the higher acuity patients are seen before they die while the lower acuity patients are seen in another area before they leave!

8.3 Infrastructure

Triage is a process, not a place, but for the most part will need an area for triage of those patients not sent directly through to resus or majors. This area should allow for privacy and be set up in order to perform the vital signs for the TEMS, additional investigations and tasks.

The room should preferably allow for one-way flow of patients from the waiting room into a subwaiting area in the area to which they have been streamed. If not possible, some sort of demarcation of the area or patients should be made in order to separate those already triaged from those who have not been assessed by the triage officer.

8.4 Alignment of staff to temporal flow of patients

It is important that triage, the tool to ensure that patients are assessed timeously, does not become the bottleneck in the system. Importantly, if the doctor is ready for the next patient, but cannot see them as they are "first getting triaged", the object is being defeated!

More often the bottleneck occurs due to failure to align staff with the flow of patients. For the most part, it is possible to predict the times of day when the flow of patients into the EC is heavier (typical "saddle-shaped" curve). Staffing for triage needs to reflect this flow.

There should also be some sort of plan in place to deal with unexpected influx of patients: each facility should have an upper limit of patients that they are willing to have waiting for triage, over which a contingency plan needs to be activated: eg a regional hospital has an agreement that if there are five or more patients needing triage, a nurse is called from majors to help triage the patients until the level is back down to less than five waiting.

IMMEDIATE TASKS	
Resus	Healthcare Professional (SHCP)
1	P for analgesia initiation
2	P for analgesia initiation
3	within 3 hours, cool the burnt area
4	newrap or clean dry sheet
5	cover clothing & discuss with SHCP
6	cover with blankets if available. Refer to SHCP
7	Resus and advise caregiver to give ORT by cup
8	Resus or assessment
9	urine bag / older child - urine container
10	P for analgesia initiation
11	P for analgesia initiation
12	pad the injured limb with a simple padded splint or
13	slings
14	to the site of trauma
15	check haemoglobin to obtain a baseline
16	check haemoglobin
17	then refer to SHCP

false;

advised to the SHCP once they have been

the bleeding area under running water.

must be referred to the SHCP for further
al rehydration.

and 20 minutes ago:

alable

minutes may

of to prevent hypothermia

in the following cases:

PAEDIATRIC PATIENTS	
PROBLEM	
1. Poisoning OR overdose	Refer to Senior HC
2. Child in pain OR inconsolable crying	Check with SHC
3. Child with a burn	Check with SHC If burn occurred Cover burn in cl Remove excess Warm the child Take to ORT, cover and spoon
4. Temperature 38.5 ° or more	Refer to SHCP
5. Temperature 35 ° or less	Refer to SHCP to younger child -
6. Diarrhoea	Check with SHC
7. Vomiting without diarrhoea but with dehydration	Immobilize affected triangular bandage
8. Presenting complaint - abdominal pain	Apply pressure to Perform finger p
9. Closed fracture	Refer to SHCP
10. Active ongoing bleeding	Perform finger p If less than 8 g/a
11. History of recent bleeding	

Assessment Questions

Clearly indicate whether the following statements are true or false.

1. A tiny baby under two months should always be referred comprehensively triaged.
☐ True ☐ False
2. A patient with an active ongoing bleed should place a dressing over the wound.
☐ True ☐ False
3. A child that is vomiting only with no diarrhoea should fit for assessment before commencing the child on a trial of oral fluids.
☐ True ☐ False

Choose the correct answer:

4. In a child with a burn injury from boiling oil that occurred 30 minutes ago, which of the following is correct?
(a) The burnt area may be covered in cling wrap if available
(b) Cool running tap water over the burnt area for 30 minutes to limit the extent of tissue damage
(c) The burnt area should be wrapped in a clean sheet
(d) a and b
(e) a,b and c
5. The senior healthcare professional should be consulted if a child presents with:
(a) Poison ingestion and overdose
(b) A severely malnourished child
(c) Vomiting only with dehydration
(d) a and c
(e) a, b and c

gency care—whether on the roadside or in the hospital—will help to save lives and reduce the need for costly hospital care by a panel of experts in Emergency Medicine. It has been shown to improve patient outcomes. However, attention needs to be paid to the training of paramedics, and the SATG recommends the use of a paramedic group.

y (MPhil) with 700 public sector patients, back following publication in four major

fect, that is why your feedback is so
at keeping the tool accurate and

waiting times in a South African emergency
care scale (adult version) provides reliable acuity
10.1016/j.hien.2011.08.002.
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Triage is an essential first step in efficient and effective emergency care in the public or private hospital arena. A robust triage tool for the South African Triage Scale has been derived from the literature (doctors, nurses and paramedics), and is scientifically validated. The triage tool can be used by doctors, nurses and paramedics waiting times and make the emergency centre run more smoothly. The triage tool has been validated as part of a Masters in Philosophy (MPhil) with 2000 private sector patients and a PhD. Feedback from the private sector has been used to improve the triage tool. Journals has contributed to the process.

This is Edition 3 and we accept that the tool may not be perfect. It is important. In addition, there will be ongoing research aimed at improving the triage tool. If necessary, subsequent editions will follow.

Online resources:

For further information please visit

www.emssa.org.za/sats

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Health